

Cybersecurity of Unmanned Aerial Vehicles: A Survey

Zhenhua Yu ^{ID} and **Zhuolin Wang** ^{ID}, Xi'an University of Science and Technology, Xi'an 710054, China

Jiahao Yu ^{ID} and **Dahai Liu** ^{ID}, Embry-Riddle Aeronautical University, Daytona Beach, FL 32114 USA

Houbing Herbert Song ^{ID}, University of Maryland, Baltimore, MD 21250 USA

Zhiwu Li ^{ID}, Macau University of Science and Technology, Taipa, Macau SAR 999078, China

Unmanned aerial vehicles (UAVs) are unmanned aircrafts operated by radio remote control and programmed control equipment. Due to their small size, low cost, and high flexibility, UAVs are widely used in military and civilian fields, such as geological detection, film shooting, traffic control, homeland security, and reconnaissance in battlefield. With the rapid increase of UAVs and the success of their related technologies, they also face many security problems, such as jamming, man-in-the-middle, and false message injection attacks. The security of UAVs has become a hot research topic. However, most UAV-related surveys are analyzed and summarized from a single perspective, there are few surveys about UAV cybersecurity. Hence, we believe that a categorized and comprehensive survey of UAV cybersecurity is urgently needed. In this tutorial, we present a comprehensive survey on UAV cybersecurity, from three aspects of the

physical domain, cyber domain, and cyber-physical domain. Specifically, in the physical domain, we review attacks that directly affect UAVs' fuselages or their significant components and discuss the solutions to these attacks. We survey the cyber threats against UAVs and the corresponding defense or detection methods in the cyber domain. One highlight is the security threat for UAV intelligent systems and the corresponding defense methods. Finally, in the cyber-physical domain, we summarize cyber-physical attacks such as GPS spoofing, these attacks are easy to destroy the chips and sensors of UAVs. In conclusion, understanding these attacks and their associated defense mechanisms will help pave a path towards the cybersecurity of UAVs. In addition, future research directions and problems to be solved are discussed at the end of this survey. With this comprehensive literature review, we set out to provide a novel insight into the current state of cybersecurity of UAVs.

Authors' current addresses: Zhenhua Yu and Zhuolin Wang are with the College of Computer Science and Technology, Xi'an University of Science and Technology, Xi'an 710054, China (e-mail: zhenhuayu@xust.edu.cn; 19308207002@stu.xust.edu.cn). Houbing Herbert Song is with the Department of Information Systems, University of Maryland, Baltimore County (UMBC), Baltimore, MD 21250 USA (e-mail: h.song@ieee.org; songh@umbc.edu). Jiahao Yu and Dahai Liu are with the College of Aviation, Embry-Riddle Aeronautical University, Daytona Beach, FL 32114 USA (e-mail: YUJ10@my.erau.edu; dahai.liu@erau.edu). Zhiwu Li is with the Institute of Systems Engineering, Macau University of Science and Technology, Taipa, Macau SAR 999078, China (e-mail: zwli@must.edu.mo). (Corresponding authors: Houbing Song; Zhiwu Li.)

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INTRODUCTION

Unmanned aerial vehicles (UAVs) have initially been developed for military applications such as air-to-ground combats. Since the 1920s, the ultimate goal of UAVs has been shifted to the reduction of casualties and property losses. As a way of obtaining remote sensing data, UAV reconnaissance has been widely used in military reconnaissance, target tracking, damage assessment, etc., which has produced enormous benefits [1].

UAVs have also been gradually integrated into civilian fields. Civilian UAVs generally include industrial and consumer UAVs. The former are applied in many fields, such as agriculture, geography, power, police, and city management [2]. The latter are mainly used for aerial photography; their performance indicators focus on shooting effects and operability. Figure 1 illustrates the current



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mainstream UAVs, including unmanned helicopter, fixed-wing UAVs, and multirotor UAVs [3].

The rapid development of UAV-related technologies has made unmanned systems more efficient and cost-effective. However, along with these technologies, UAVs face external and internal security threats unforeseen in the past. From the external perspective, UAVs may be shot down by destructive weapons. On the other hand, the network and communication data links of UAVs will be invaded or attacked, causing the loss or tampering of sensitive data in UAVs. For instance, UAVs communicate with a control station through wireless networks, and attackers can attack navigation systems through global positioning systems (GPS) spoofing [4], or attack networks through jamming attacks, thereby exerting an interference effect on UAVs [5]. Once cyber attacks occur, they may make UAVs unable to work as expected, even causing casualties and property losses. Consequently, the security of UAVs has become a hot topic.

The cybersecurity of UAVs has been extensively studied in many papers. However, the existing surveys do not comprehensively and categorically summarize the various attack and defense methods faced by UAVs. Meanwhile, most surveys are for the single characteristics of UAVs summarized, which rarely mention security of UAVs. For example, Sharma et al. [6] investigated task modules, antenna, resources processing

platforms, and network architectures, and summarize the challenges and future research direction of UAV communications. Wang et al. [7] summarized UAV networks from the perspective of cyber-physical systems (CPSs) [8] and divide them into three levels: cell level, system level, and system of system level, to study what contributions these components can make to the system performance. The following are some surveys related to the information security of UAVs.

Yaacoub et al. [8] proposed a new idea to find and deal with the security threats of UAVs. They use UAVs for cyber attacks to identify new threats, and then develop strategies to defense them. This article mainly analyzes the security vulnerabilities of UAVs. However, it does not comprehensively summarize all kinds of security threats to UAVs.

Altawy et al. [9] mainly summarized the various attacks on civilian UAVs. In particular, they also identified some of the physical and cyber threats to UAVs and discuss the security features required for their critical operating environment. The advantage of this review is that it summarizes the security threats faced by civilian UAVs, but does not to respond to the intelligent systems security threats that UAVs may face. Meanwhile, this article does not consider the possible security threats to military UAVs.

Therefore, a survey that can comprehensively classify and summarize UAV cybersecurity threats is urgently

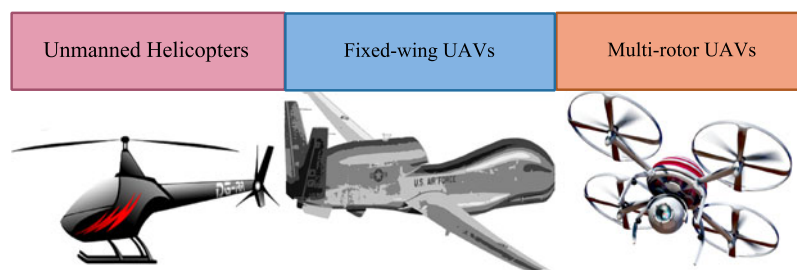


Figure 1.

Current mainstream UAVs.

needed. Also, due to the widespread use of artificial intelligence (AI), the addition of security threats related to UAV intelligent systems should be considered.

In this article, three domains (physical domain, cyber domain, and cyber-physical domain) involved in CPS are proposed as the idea of summary, and various security threats faced by UAVs and corresponding defense methods are comprehensively and clearly summarized. The contribution of this article can be summarized as the following points:

- 1) In the physical domain, this article discusses attacks on the fuselages and crucial components of UAVs. It also summarizes detection methods of the Hardware Trojans and defense methods of human factor attacks.
- 2) In the cyber domain, the security threats and countermeasures in the cyber domain are summarized from three parts: traditional software security threats, intelligent system security threats, and network threats. In particular, we consider that UAVs may also face AI security threats. Therefore, the security threats of intelligent systems are added to achieve a systematic and comprehensive review.
- 3) In the cyber-physical domain, we mainly review security threats coming from the sensors carried by UAVs and their corresponding solutions.

The main body of this survey is sections “UAVs Security Threats,” “Defense Against Various Attacks of UAVS,” and “Future Research Challenges,” which are 3D structures (physical domain, cyber domain, and cyber-physical domain). Section “Understanding of UAVs” introduces the components and applications of UAVs. Section “UAVs Security Threats” discusses the concept of UAV network security and presents a taxonomy of various

attacks on UAVs from three domains. Section “Defense Against Various Attacks of UAVS” touches upon the defense or detection methods of UAVs against a variety of attacks. In sections “UAVs Security Threats” and “Defense Against Various Attacks of UAVS,” we discuss intelligent system security threats of UAVs and corresponding defense methods. Sections “Future Research Challenges” and “Conclusion” point out future research directions about the security of UAVs and conclude this article, respectively.

UNDERSTANDING OF UAVS

COMPONENTS OF UAVS

As shown in Figure 2, a typical UAV system usually includes a flight system, a set of communication data links, and a ground control system (GCS) [9].

The flight system includes an airframe, a power system, a navigation system, a communication system, and a flight control system. The airframe is the supporting platform of a UAV, which can support all the equipment to fly into the sky. The power system provides a UAV with long-lasting endurance. The navigation system measures and calculates the position, speed, and flight attitude of a UAV with a reference coordinate system. Also, the navigation system guides a UAV to fly according to the designated route, equivalent to a navigator in manned aircraft systems. The navigation system works with other sensors, such as barometers and gyroscopes [9]. The status data of a UAV can be measured and transmitted to a GCS for control signals analysis. The communication system provides a data link between the control station system and a UAV. The flight control system, also called flight management and control

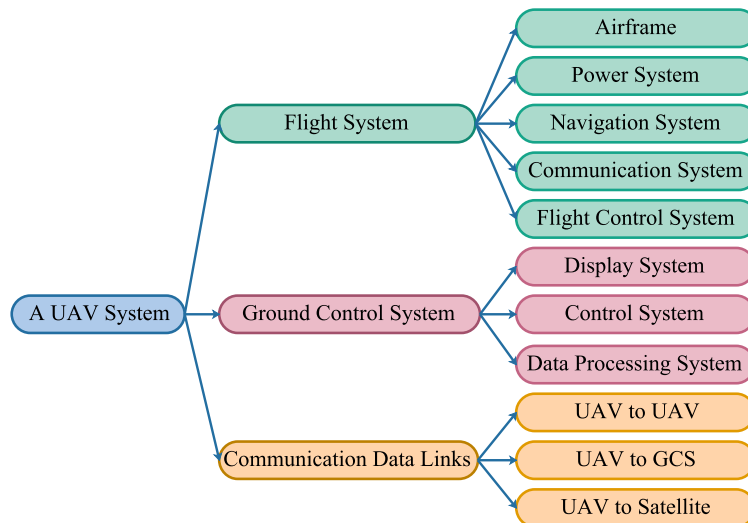


Figure 2.
A UAV system.

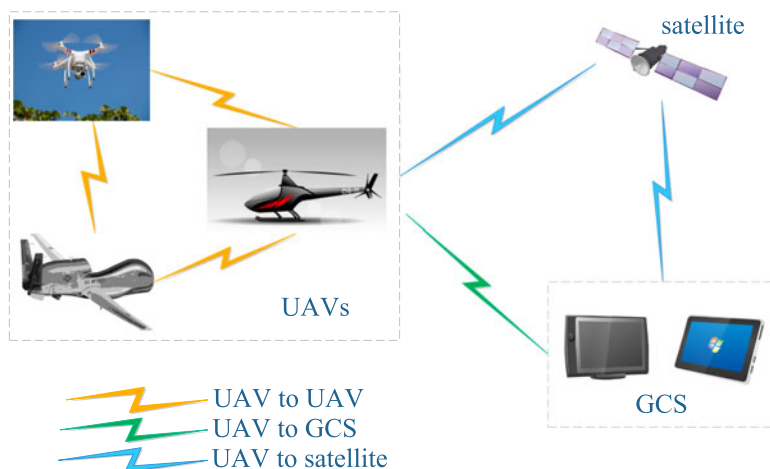


Figure 3.

Communication systems between UAVs, GCSs, and satellites.

systems, is equivalent to the “heart” part of UAV systems. The main task of the control system is to maintain the stability of the altitude and flight paths of an aircraft.

The transmission of control information between UAVs and GCSs mainly relies on the communication system, which allows data exchange [10]. The relationship of the three communication links is illustrated in Figure 3.

A GCS is a vital part of a UAV system, including a display system, a control system, and a data processing system. The display system is mainly used to expose the flight status of a UAV in real time. It monitors these status through parameters such as the number of satellites locked by GPS receivers on a UAV, angles of a gimbal, and a camera exposure. The control system mainly controls the flight of the aircraft through the course planning software to plan the flight path, or manages the flight of the UAV manually through the console. The primary functions of the data processing system are to analyze and process data transmitted by UAVs and transmit flight parameters of UAVs to control systems in real time. Then, the system shares the data to display systems through data transmission links [9].

In order to keep UAV fly safely, UAVs also consist of support equipment, which can be used for maintenance, recovery, fault diagnosis, etc.

APPLICATION AREAS OF UAVS

According to practical requirements, UAVs are divided into civilian and military ones.

CIVILIAN UAVS

They are usually small and easy to operate. Their specific application fields are presented in Figure 4.

- *Agriculture:* UAVs use remote sensing technologies to perform precise farmland irrigation [11], analyze soil and moisture of farmlands [12], and also use the remote sensing image to manage farmlands [13]. Image analysis technologies of UAVs can also be employed to classify crops [14]. Subba et al. introduced a kind of UAV that can realize semiautomatic pesticide spraying, thus reducing the harm of pesticides on humans, which reduces the farming time, and manual labor of farmers [15].
- *Geography:* UAVs can gather terrain data, such as capturing dense points in the images of forest areas [16], collecting terrain data of monuments [17], and using UAV photogrammetry technology to obtain the parameters of mine collapse in a short period [18].
- *Police Force:* UAVs are applied to monitor road traffic [19]. When encountering emergencies, such as flash floods, tsunamis, and earthquake disasters, high-definition cameras can be used to take aerial photos of the disaster-stricken areas [20] and promptly transmit the latest images to control stations. It is crucial to transmitting real-time images for the rescue work [21].
- *Urban Management:* In a project construction site, UAVs can be used by a project manager to monitor the progress of the project, which contributes to a higher efficiency. UAVs can be employed to check critical ground features, such as signal towers [22]. Smart UAVs are expected to provide new opportunities in different applications, especially in the civilian field, making urban management more convenient and efficient [23].

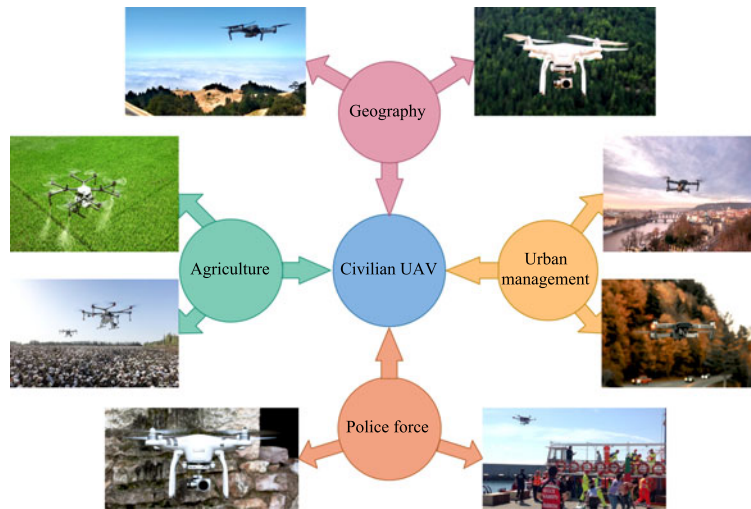


Figure 4.
Classification of civilian UAVs.

MILITARY UAVS

The application fields of military UAVs are illustrated in Figure 5.

- *Reconnaissance*: UAVs are equipped with a variety of sensors, such as photoelectric and radar, to get a comprehensive reconnaissance capability. UAVs perform reconnaissance missions by planning a route to navigate a predetermined route, or randomly exploring without prior knowledge [24]. They can also be used for camp security missions [25].
- *Military Strikes*: These types of UAVs can attack targets and engage in combat missions. They can reduce casualties and strengthen the combat capabilities of troops [26]. Also, they can be applied to various tactical tasks [27].
- *Homeland Security*: When law enforcement and border control monitoring are difficult, UAVs are particularly important for border protection. For example, in the case of Libya which has become a transit bridge for illegal immigrants, mostly from South Africa to Europe, high-definition UAVs are used to monitor the border situation [28].

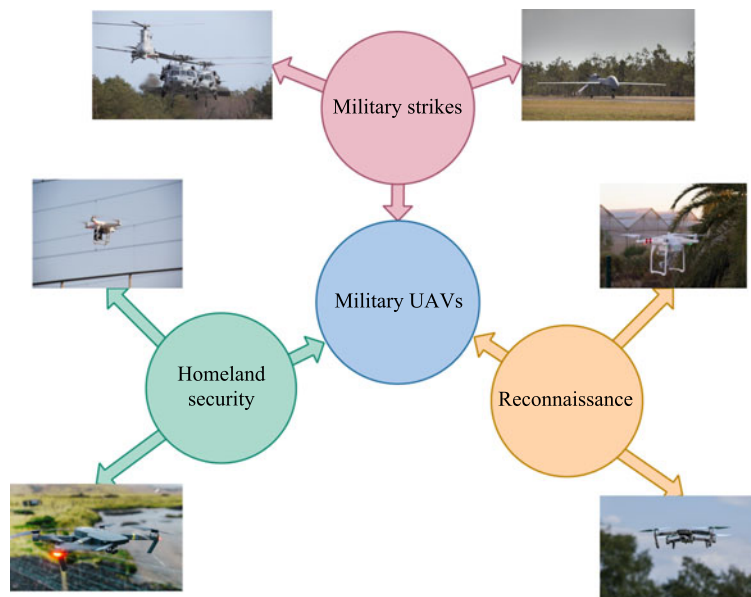


Figure 5.
Classification of military UAVs.

UAVS SECURITY THREATS

The rapid development and success of UAV-related technologies have brought security issues to UAVs. In this article, we study the security threats of UAVs from three domains, as shown in Figure 6. In the physical domain, a major security problem is that destructive weapons may maliciously damage UAVs. In the cyber domain, the security problems (such as injection attacks, jamming attacks, and hijacking attacks) are mainly caused by attacks on communication networks, which leads to data loss or jeopardizing UAV flight processes. Eventually, UAVs are out of control or even destroyed. In the cyber-physical domain, security issues are through cyber attacks [29], [30] that impact physical environments or devices. The flight systems, data links, ground control stations, and other support equipment may face various security threats.

UAVS SECURITY THREATS IN PHYSICAL DOMAIN

The attacks in the physical domain of UAVs generally aim at the fuselage of UAVs and their significant components. These attacks target at small, low-flying UAVs, such as military UAVs that carry out border surveillance or reconnaissance missions. They could be interfered by cyber attacks and guided into enemy territory. Also, destructive weapons can shoot them down. These attacks fall into the following categories.

- *Physical Hard-Kill and Soft-Kill Attacks*: Physical hard-kill attacks are the direct destruction of UAVs. The attacks may affect the fuselage and other vital components, such as a UAV's wings. At present, the raids are mainly conducted by projectile kinetic energy or directed weapons.

Physical soft-kill attacks affect UAVs' expected flight or affect their body parts to make them disabled. These methods include high-speed water flow, sturdy clusters of objects, dense foam, etc. However, the applications of these attacks are limited, and entail essential high-quality logistics support; these shortcomings weaken the actual deployment capabilities of physical soft-kill attacks [31].

- *Malicious Hardware Attacks*: Malicious hardware attacks are deliberately tampering with UAVs' hardware design so that they operate in a manner that is not specified by the designer. This threat may cause malfunction and confidential data leakage of UAVs. For instance, an attacker could implant malicious hardware into UAVs to monitor them [32].

Malicious hardware attacks may also be a chip on UAVs, which are deliberately designed hardware Trojans to disable specific security mechanisms of UAVs. Once this malicious hardware is triggered, it could

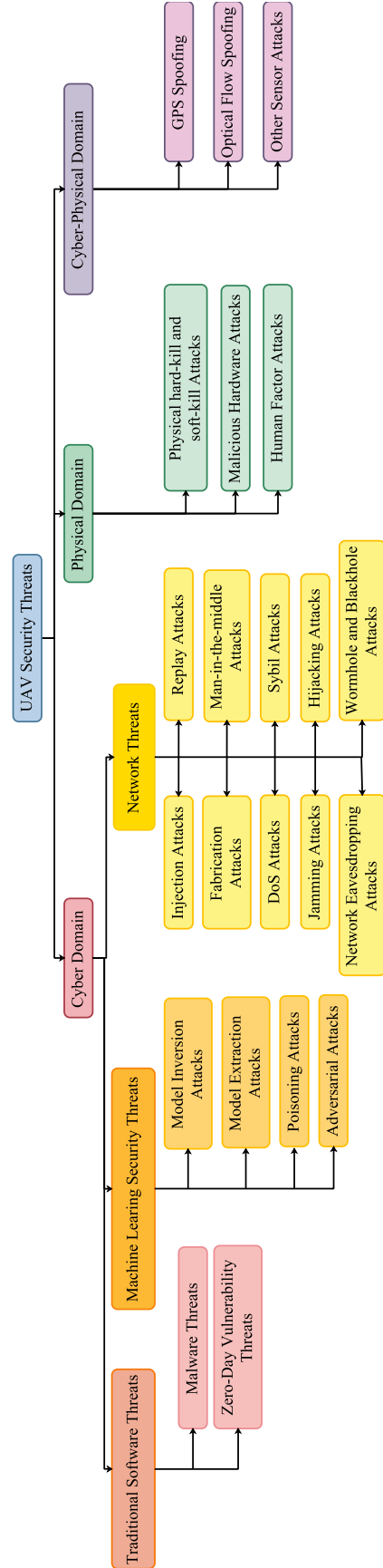


Figure 6. Overall structure diagram of UAVs attacks.

result in disastrous consequences [33]. Two security experts from Cambridge discovered the hardware Trojan on the new Boeing 787 airliner. Hardware Trojans have also been found in aircraft equipped with Actel ProASIC chips, which could allow passengers to use aircraft entertainment systems to take over avionics and control planes [34]. If terrorists use the virus to control a flight, the consequences would be disastrous. Although the chips of UAVs are different from those of the general airliners, this hardware Trojan virus can also be used in UAV chips.

- *Human Factor Attacks:* UAVs flying at line-of-sight are the targets of human factor attacks such as destruction and theft. These attacks can be encountered in many ways, from a simple dart gun to an anti-UAV rifle. Specifically, there is an anti-UAV rifle that uses radio pulses to disable a UAV within 1,300 feet without damaging its body, but disrupting its communication data link. The affected UAV immediately enforces the fail-safe protocol and hovers on the ground nearby [33]. This method not only destroys the communication data link but also increases the risk of steal.

UAVs teams that use a same operation could probably increase the chance of unintentional collision between UAVs. These inadvertent collisions may cause UAVs to adopt induction and avoidance mechanisms [35]. Therefore, attackers could attack the UAVs' avoidance mechanisms, abusing them by flying another UAV at close range, and forcing the wrapped helicopter to flee until its battery runs out and launch a denial-of-service (DoS) attack.

- *Summary Attacks in Physical Domain:* In recent years, the security threats UAVs in the physical domain have emerged. While most UAVs encounter attacks in the cyber domain, civilian UAVs also encounter some attacks in the physical domain.

First of all, most direct attacks are physical hard-kill methods. UAVs flying at low altitude can be shot down through a projectile kinetic energy weapon or a directed energy weapon. The defense method against these attacks is by relying on an operator to pay attention to the movement of UAVs to prevent them from being destroyed by attackers. Second, UAVs may also be subjected to physical soft-kill attacks. These attacks increase flight difficulties of UAVs through several methods, such as solid clusters of objects like particles, dense foam, etc. Finally, human factor attacks can also pose a security threat to UAVs, such as destroy UAVs or steal them. Among the defense methods, the use of electronic anti-theft locks is against theft.

UAVS SECURITY THREATS IN CYBER DOMAIN

Wireless communications make UAVs vulnerable to a variety of attacks in the cyber domain. These attacks can have a huge negative impact. Since military UAVs mainly carry out secret missions such as strike, rescue, and reconnaissance, once these UAVs attacked by cyber threats, it will cause huge military losses. Civilian UAVs are vulnerable to cyber threats and could be attacked by hackers or attackers. Therefore, both military and civilian UAVs are vulnerable to attacks in the cyber domain and need constant attention. As shown in Figure 6, the threats in the cyber domain are classified into traditional software threats, machine learning security threats, and network threats.

TRADITIONAL SOFTWARE THREATS

Exhibited in Table 1, traditional software threats against UAVs are mainly caused by malware and security vulnerabilities. Such attacks mainly destroy the confidentiality of information security for UAVs.

- *Malware Threats:* Malware is a major cause of security threats to UAVs' software. The well-known "rogue software" tracks online behavior and provides the personal information to "hidden" market interest groups. Moreover, they can pop out advertisements through the malware. It can be divided into "spyware," "malware," and "deceptive adware." Malware implantation is the main attack method for attackers to invade and control UAVs [36].

Table 1.

Summary of Traditional Software Threats	
Types	Description
Malware	This attack obtains user information, and maliciously tamper with the registry information [34].
Maldrone	This malware acts as a communication agent in the communication link and injects false data to control UAVs [30].
Keylogger	This malware records every keyboard operation of the UAV to monitor confidential information [35].
Zero-Day Vulnerability	This vulnerability is a malicious program that almost simultaneously appears in the software as a security vulnerability [38].

Malware can obtain the information and privacy of users, maliciously tamper with registry information, and potentially manipulate users' devices remotely without their permission. If malware is deliberately inserted into military UAVs, these UAVs could be hijacked to disrupt their missions and potentially steal sensitive information. In 2018, Vanunu et al. [37] discovered a vulnerability in DJI UAVs, which was generated by DJI forum access. When a user logs on the DJI forum and unwittingly clicks on the malware planted, an attacker will obtain the login credentials, access other DJI online assets, and then gain the live camera view of UAVs from the perspective of a map.

In 2011, Keyloggers infected computer types of equipment to control UAVs in the United States. They are spread between a secure network and a public network through a hard disk or a removable device [39]. The original keylogger is used to monitor a user's keyboard operations, but now it can track and monitor almost everything, from computer screens to internet usage. Also, the resources of keyloggers can be shared with legitimate programs. From this point, attackers could use this feature to attack the control software of UAVs [38]. In the same year, Iran claimed that they had captured and decoded intelligence from UAV systems for military purposes, which manifests the actual threat and intentions of enemies against government UAV systems [39].

Among viruses infesting UAVs, there is a malware called Maldrone. Once this software is installed on a UAV, a botmaster may activate it, and a backdoor can be opened to receive commands. Maldrone acts as a proxy for communication between a UAV flight controller and sensors, injecting the values required for communication. In other words, the malware can land any infected UAV at a location chosen by an attacker [33].

- **Zero-Day Vulnerability Threats:** Zero-day vulnerability is a kind of security vulnerability that is not disclosed to vendors or developers. Since it is difficult for developers and security experts to find all security vulnerabilities, they expect them to exist and spend a lot of efforts to find security flaws [40].

In [41], Hooper et al. developed a security attack threat model by conducting penetration tests on Parrot Bebop UAVs and discover three zero-day vulnerabilities. They find that vulnerabilities discovered through DoS attacks, buffer overflow, and Address Resolution Protocol cache poisoning can disrupt the flight behavior of Bebop UAVs. From this, they infer that UAVs may be vulnerable to several other common security attacks.

In Taiwan, a LILIN company monitors multiple zero-day vulnerabilities through its digital video recorder systems. However, hackers use these

vulnerabilities to attack devices infected with viruses or those vulnerable to distributed denial of service attacks. Qihoo 360's Netlab team discovers these zero-day vulnerabilities. It said that in 2019, different attack organizations used these zero-day vulnerabilities to spread certain botnets [41]. Javaid et al. [42] pointed out the possibility of attacking UAV systems through zero-day vulnerabilities.

MACHINE LEARNING SECURITY THREATS

As a typical system for future operations, intelligent unmanned systems are highly informative and smart electronic information equipment. Machine learning technologies are widely used in various UAVs, such as target recognition, tracking, intelligent control, path planning, etc. However, they are facing many security threats [43]. This section analyzes machine learning security threats of UAVs, which are summarized in Table 2.

- **Model Inversion Attacks:** In the early learning, there is only one information flow between the training dataset and the training model—from the dataset to the model. However, it turns out that there is also a reverse information flow, recovering dataset information from model information, which is called a model inversion attack.

Model inversion attacks use machine learning models provided by some application program interfaces (APIs) to obtain information models about the system. It then reversely inverts this information (as exhibited in Figure 7) to obtain private

Table 2.

Summary of Security Threats to Machine Learning	
Types	Description
Model Inversion Attacks	This attack targets the defect that machine learning implicitly memorizes some over-fitted training data [44].
Model Extraction Attacks	This attack refers to restoring the original model by receiving the returned model data [47].
Poisoning Attacks	This attack refers to maliciously tampering with model data [44], [51].
Adversarial Attacks	This attack refers to a malicious attack that weakly disturbs standard samples and completely invalidates the DNN [54].

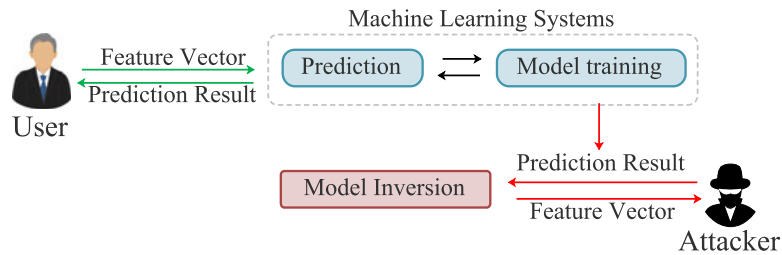


Figure 7.
Model inversion attacks.

information in the training dataset, such as user record information and biometric data [44].

Based on previous works, Fredrikson et al. [44] studied model inversion attacks on a decision tree algorithm, and realize white-box attacks and black-box attacks on the decision tree algorithm. They further used a gradient descent algorithm to reconstruct the corresponding face image, and continuously call the model to obtain the output, and ultimately obtain the face image data for training. Ateniese et al. [45] showed that the statistical properties of training data can be inferred from machine learning models.

In short, model inversion attacks steal the private information of members in the training dataset and damage the dataset owner's commercial value. There are two reasons for this situation: one is insufficient privacy protection, such as information leakage [46], and the other is insecure algorithms [47].

- **Model Extraction Attacks:** Model extraction attacks occur on trained models and are mainly used to steal model parameters and illegally obtain models. These attacks refer to the fact that attackers send a large amount of prediction data to models through the API provided by the deep learning system. They then receive the class label and confidence coefficient returned by the model, calculate the model parameters, and thus restore the original model.

The training process of a model requires the collection of a large number of datasets, a large amount of time, and powerful computing ability. Once the model is extracted and abused, the model's privacy can be destroyed and bring substantial economic losses to the model owner [49]. Wang and Gong [48] in their work attempted to steal hyper-parameters on the premise of known model algorithms and training data.

The purpose of model extraction attacks can be divided into three categories: obtaining free use rights of the model, stealing the privacy of training data, and bypassing security detection. Tramer et al. [49] proposed to launch attacks against current mainstream machine learning models through predictive APIs to extract target machine learning models and obtain

free usage rights. Meanwhile, an increasing number of application scenarios use machine learning models to detect malicious behaviors [50]. After extracting the target model, attackers can construct corresponding adversarial samples based on relevant knowledge to bypass security detection [51].

- **Poisoning Attacks:** Poisoning attacks are implemented by attackers maliciously altering the input of users to mislead models [52]. They mainly refer to the manipulation of machine learning model predictions by attacking a training dataset or algorithm during a training process. Methods of attacking a training dataset include polluting source data, adding malicious samples to the training dataset, modifying some labels in the training dataset, and deleting some original samples in the training dataset.

As illustrated in Figure 8, poisoning attacks produce a polluted model after polluting the training data. The attacking algorithm takes the advantage of an insecure feature selection method or a training process algorithm's weakness. Eventually, the poisoning attacks open a backdoor for the attacker in the generated model, to bias the prediction of the model to an expected direction [53].

In the process of data poisoning, false labels or partial data are intentionally processed and added to a training dataset, reducing the overall quality. According to related investigations, there are two sources for data poisoning attacks, namely, incorrectly labeled data and specific obfuscated data. Xiao et al. [54] adjusted the training set to attack support vector machines by flipping labels. They proposed an optimized framework to find label flips to maximize the classification error, thereby reducing the accuracy of the classifier.

Deep neural networks (DNNs) effectively improve the performance of image, speech, and pattern recognition. However, poisoning attacks are serious threats to the security of DNNs. The attacks reduce the accuracy of DNNs by adding malicious training data during the training process. In some cases, it is necessary to delete a specific accuracy level from the model. For example, if an attacker does not allow selective



Figure 8.
Poisoning attacks.

identification of nuclear facilities, UAVs will be deliberately prevented from correctly identifying nuclear-related facilities [55].

- *Adversarial Attacks:* Existing studies have shown that DNNs are vulnerable to adversarial samples and adversarial attacks in deep learning. Adversarial attacks mean that an attacker adds small interference to a specific dataset design during the model testing phase. Thus, an attacker obtains antagonistic samples, making the DNNs' high recognition rate completely invalid, and causing them to output the misjudged result [56]. The main attack methods include the white-box attacks and the black-box attacks.

White-box attacks mean that an attacker can access a target model and grasp parameters used in the model. While constructing adversarial samples, attackers can interact with the target model [57].

Black-box attacks mean that an attacker does not know the internal details of a target model. An attacker continuously interacts with the target model to train a model that is infinitely close to the target model. Then this approximate model is used to generate adversarial samples [51].

NETWORK THREATS

UAV networks are an ad-hoc network, that is mainly composed of base stations and a limited number of UAVs. During the operation of this network, GCSs control and monitor UAVs based on the information sent from the onboard equipment of UAVs through a wireless channel. However, these networks are essentially open and lack physical infrastructure. UAVs may be subject to cyber threats such as jamming or eavesdropping attacks [58].

- *Injection Attacks:* Injection attacks are a kind of integrity attacks, which means that attackers modify the sensitive information of UAVs by injecting the wrong data, so as to make false data appear in data streams and transmit to GCSs [59]. Figure 9 presents a summary of various injection attacks.

In a false communication data injection attack, an attacker chooses a weaker communication channel, blocks legitimate data, and transmits erroneous data to a receiver [59]. The attack can be completed through sensor communication links or command and control links. Complex false communication data injection attacks are not easy to detect [58].

False database data injection attacks mean that an attacker attacks a database of UAVs, which may damage the database or even eradicate it. The structured query language (SQL) injection attack is a shared database threat by which attackers would use the vulnerability to destroy a database [59].

Command injection attacks are executed by an attacker extending the injection function on the basic software of UAV systems, through some commands to perform malicious acts or gain unauthorized access to a UAV system [59]. For example, a controller area network bus code injection can not only control a UAV system physically, but also inject code to manipulate a remote terminal unit [60].

Code injection attacks are ones from which most software of UAV systems have suffered. The main goal of these attacks is to manipulate the behaviors of the original software. This type of attacks takes the advantage of the design flaws in software and databases, causing a severe threat to a UAV system [59].

Sensor false data injection attacks are to inject a set of incorrect sensor data into a flight controller, thereby destroying the stability and security of UAVs. This

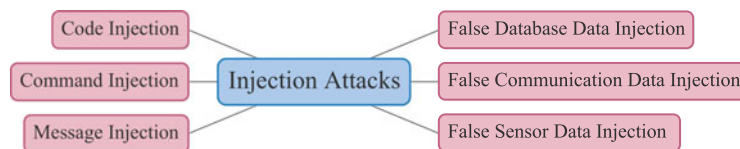


Figure 9.
Classification of injection attacks.

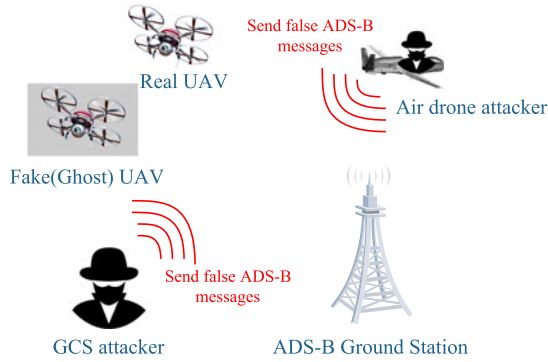


Figure 10.
Message injection attacks.

type of attacks is mainly used to control sensors affected by external influences, such as radar, infrared sensors, and photoelectric sensors. Once UAVs are subjected to the injection attacks, they may cause UAVs to lose control and crash [32].

In a message injection attack, an attacker injects fake legal messages into automatic correlation surveillance broadcasting (ADS-B) networks [61]. There are two general message injection attacks, namely ground station target ghost injection and aircraft target ghost injection. As shown in Figure 10, attackers perform ground station target ghost injection. The attributes of a series of false ADS-B messages distributed by two attackers (ground and air attackers) are the same as those of legal ADS-B messages. Thus, GCSs may treat fake UAVs as real ones [62].

- *Fabrication Attacks:* Fabrication attacks belong to integrity attacks too, which are often combined with other attacks to launch an attack on UAVs. The attack process is that an attacker uses a deceptive identity to obtain the confidential information of UAVs, and feeds incorrect data back to GCSs [59]. GCSs may suspect that this UAV as an attacker,

which increases the misjudgment rate greatly. In other words, GCSs may misjudge a real UAV as a fake one [63]. For example, when targeting medical devices [64], an evil twin attack is a typical fabrication attack, which is mainly executed in two stages. First, an attacker pretends to be a UAV and creates a new access point (AP), and uses deauthentication to attack a legitimate AP. Second, the attacker pretends to be a legitimate AP by launching a fake AP, thereby forcing the client to reauthenticate with the fake UAV acting as a legitimate AP [65].

Nondestructive soft kill methods can be regarded as a fabrication attack, which falsifies or destroys information, control signals, and control links of UAVs [31]. A typical fabrication attack model is indicated in Figure 11. If a real operator controls a UAV through remote control signals, an attacker monitors and recognizes the real remote control signals, and then sends false remote-control signals to the UAV to block the real operator's control.

UAVs equipped with ADS-B systems can effectively avoid collisions with other manned or unmanned aircraft. However, these ADS-B signals are particularly susceptible to interference or fabrication by false signals. If an attacker continues to provide malicious ADS-B signals to a UAV, it may deviate from its route and be guided to a false destination [33].

- *DoS Attacks:* DoS attacks can cause the failure of servers to provide services. An attacker attacks a medical device client in a medical field through deauthentication attacks [66]. It would cause disconnection among all devices connected to the client [67]. DoS attacks on UAVs refer to exhausting the energy of UAVs or interfering with the network and routing protocol [59], to jam the normal operation of UAVs, or even paralyze UAVs [68].

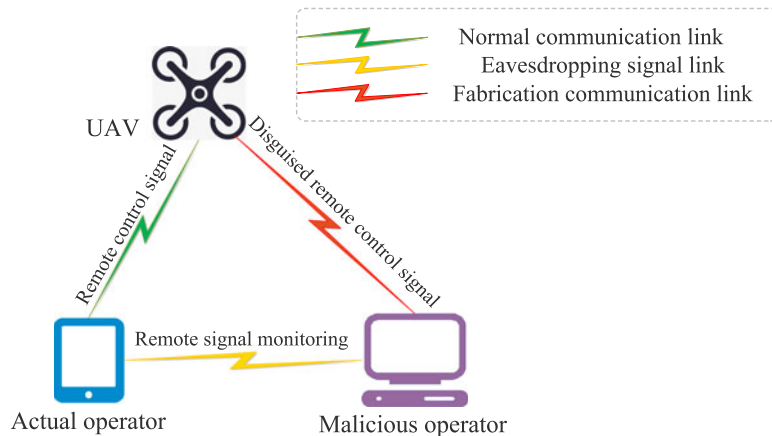


Figure 11.
Fabrication attacks.

Distributed denial of service (DDoS) attacks are a typical type of DoS. DDoS attacks refer to multiple attackers in different locations, simultaneously launching attacks on one or several targets. An attacker controls various devices located in other places to attack the target device simultaneously. Javaid et al. [69] proposed threat models, including DDoS and jamming attacks. Javaid et al. [70] used a UAVSim simulation test platform to evaluate the impact of DDoS attacks on UAV networks. They used the traditional method of transmitting deceptive packets to a single host. In this method, all attacking hosts behave the same as regular UAV hosts, and are assigned the same IP addresses range, making it indistinguishable from other trusted UAVs.

Commercial UAVs are mostly small UAVs, whose functions include the delivery of goods and services. If attackers can access to flight controllers, UAV systems could be compromised, and subjected to DoS attacks. In plain words, these attackers can manipulate flight control commands including stop commands. When UAVs are running, they can illegally invoke the stop command [33].

In addition, DoS attacks against CPSs mainly act on network control systems. UAVs as an application of CPSs, their control systems would also suffer the attacks. Many researchers use mathematical models to analyze threats of DoS attacks on CPSs. Long et al. [71] used two queuing models to simulate the random process of packet delay jitter and loss under DoS attacks.

- *Jamming Attacks:* Jamming attacks aim to interfere with wireless communication links and communication signals through technical means. The interference to signals of UAVs is generally the interference to communication signals for UAVs, forcing the interruption of signals. In the event of a loss of signals, UAVs would typically land where it was, return to their set point or continue sailing, allowing hackers to attack accordingly. When an operator was shooting a triathlon event with a UAV in Australia, an attacker interfered with the communication frequency between ground controllers and UAVs, causing UAVs to fall on an athlete [72]. An attacker used an unencrypted wireless communication channel between a UAV and a smartphone to inject a deauthentication code, causing legitimate users to disconnect from UAV networks [58].

With the advent of software-defined radio (SDR) technology, programmable smart wireless attack devices may pose a most significant threat to UAV networks. Intelligent attack behavior can automatically perceive the status of the wireless spectrum. It can launch various UAVs attacks such as eavesdropping,

jamming, and deception, by analyzing the characteristics of the defender's behavior. Since jamming attacks have the lowest technical threshold, the lowest implementation cost, and the most direct attack effect, the use of intelligent attack behaviors to launch jamming attacks on UAVs directly weakens or prevents UAV network communication links [73].

Cooperative jamming attacks are complex jamming methods, which can also be used as a defense means of eavesdropping attack. It mainly uses other nearby UAVs to jam malicious devices. If attackers use this method to attack UAVs, it will lead to the deterioration of the quality of the transmitting signals and directly interfere with the execution of UAV missions [74].

Electromagnetic interference (EMI) attacks can be performed such as jamming, requiring a small number of resources and power. These attacks usually target communication links to degrade their performance and ultimately result in the loss of all UAVs [60].

- *Network Eavesdropping Attacks:* Network eavesdropping attacks are passive attacks. Attackers usually use malicious devices to eavesdrop on the message exchanged between UAVs and air traffic control to understand what is in the air or when a UAV appears. In this case, an attacker changes from being passive to active, chooses the appropriate time to block communication, and creates wrong environmental information by injecting fake information into the air traffic controller, such as providing false UAVs location. Then, they subsequently launch cyber attacks [59]. For instance, when a UAV performs a task, it continuously transmits its encrypted location information to a GCS through wireless communication. An attacker could monitor the wireless links and collect all the encrypted information [75].

Network eavesdropping is a management tool that monitors network status, data flow, and information transmission online. It can set a network interface to a monitoring mode and intercept the information transmitted on networks. These transmissions are between the UAVs and GCSs [76]. When users, GCSs, and UAVs communicate over wireless links, an attacker eavesdrops to the wireless links between them through the network, causing sensitive information to leak [77].

Compromising emanations is a type of eavesdropping attack through electromagnetic, optical, and acoustic emission, bypassing all security measures, such as embedded encryption. The U.S. Department of Defense (DoD) points out that compromising emanations would disclose important safety information, which could be maliciously

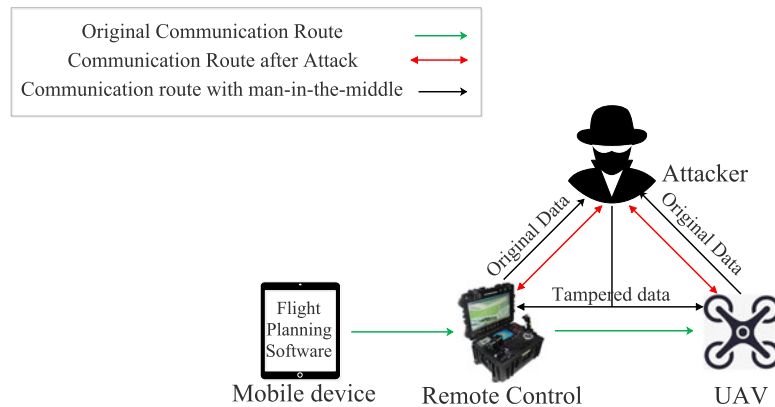


Figure 12.
Man-in-the-middle attacks.

used, posing a safety threat to UAVs. Compromising emanations could affect confidential systems and any systems targeted by adversaries, including wired and wireless devices and systems [60].

- *Man-in-the-Middle Attacks:* Man-in-the-middle attacks intercept normal network communication data, and then perform data tampering and sniffing so that both parties cannot communicate normally. Session hijack and domain name server (DNS) spoofing are two typical man-in-the-middle attacks. According to Figure 12, after receiving original data from a remote control system and UAVs, an attacker, as an intermediary, tampers with the data and sends it to the remote control system and UAVs in the same way. The attacker starts unauthorized network eavesdropping of the communication link.

Attackers can use man-in-the-middle attacks to invade pieces of UAV control software, and then remotely change various parameters of the software. When researchers investigate the vulnerability of UAVs, they use “remote AT commands” to launch a man-in-the-middle attack on the XBee 868LP chips. The commands can remotely change internal parameters of the XBee chips, thereby rerouting any traffic. They also reverse-engineer flight computers and flight planning software, and then match commands transmitted through the telemetry channel with specific functions in UAVs. Therefore, an attacker can modify existing data packets or inject new data packets to communicate with the flight computer [75].

An Australian attacker entered the air traffic control frequency via radio transmission, communicated directly with pilots and control towers posing as the control tower, and ordered aircrafts to “go around” instead of landing, leading to an Australian Federal Police investigation [60].

- *Replay Attacks:* Replay attacks destroy the integrity of UAVs. An attacker uses network monitoring or

other methods to steal authentication credentials and then re-sends the stolen authentication credentials to a receiver to achieve the purpose of deceiving UAVs. Replay attacks can also be understood as malicious or fraudulent repeated data transmission. For example, in a Byzantine replay attack, an attacker can repeat data recorded from a damaged sensor or actuator within a certain period [78].

Video replay attacks are kinds of replay attacks on UAVs. When an assailant attacks UAVs, an operation control unit of UAVs is sneaked into, and real-time videos of a camera are replaced with recorded videos, which could be used to confuse operators. When an operator inspects UAVs in an autonomous mode, it is difficult to determine whether it has been attacked by tracking all the information of UAVs. VideoJak, as an IP video security assessment attack, proposes a video playback attack on IP cameras. IP cameras are similar to cameras used in UAV systems [79].

Replay attacks have an efficient way to destroy the software of UAVs, repeatedly sending cancel authentication packages (also known as “Aircrack”). This method can work against the connection between a UAV and an operator, but its disadvantage is that it only works on specific UAVs at a short distance [31].

- *Wormhole and Blackhole Attacks:* Wormhole attacks are usually launched through the cooperation of two or more malicious UAVs. Two hostile UAVs located in different locations detour to receive messages from another channel and pass them to another malicious node through a dedicated communication channel. When UAVs are under attacks, a malicious node receives a packet UAV position and then transfers it to another remote malicious node through the tunnel, so the adjusted data packet would be sent to its neighbor nodes. The attack is mainly used as a malicious bait node between a source node and a destination

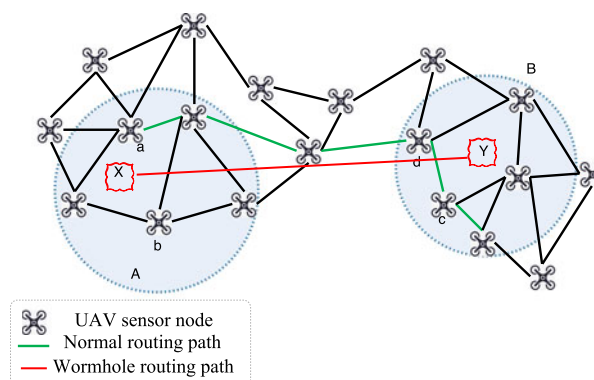


Figure 13.
Wormhole attacks.

node. When a UAV encounters a wormhole attack, the attack can make the receiver of the UAV loss data packets and change its operations [80].

As illustrated in Figure 13, UAVs in areas *A* and *B* are normal nodes. Generally, the distance from node *a* to node *b* is five hops. After two wormhole nodes *X* and *Y* join UAVs, the transmission distance between the two nodes becomes three hops. Since most ad hoc network routing protocols select paths based on jumps or distance, the routing between node *a* and node *c* no longer passes through the five-hop routing path. It seems to save transmission time; however, if these two wormhole nodes are malicious, then an aggressor can launch a series of attacks to destroy UAV networks.

Blackhole attacks are a type of DoS attacks by dropping packets. When a routing table misses entries of target UAVs, blackhole nodes can send a forged routing response after receiving routing request packets, claiming to have a short and unreached route. The primary process is that when the prefabricated RREP packet arrives at the source node, the malicious node will establish its route and discard all legitimate RREP messages transmitted from other intermediate nodes and target nodes. At this point, the malicious node successfully introduces the data stream to its destination [81]. In addition, blackhole attacks can forge RREQ packets and the number of hops in the field-source sequence number, and attack source nodes, intermediate nodes, and destination nodes to mess up the routing table [82].

Gray holes are malicious nodes interrupting data transmission in the network by transmitting incorrect routing information. Due to the unpredictable nature of the malicious nodes, gray hole attacks are also seen as an extension of blackhole attacks [81]. They interrupt the route discovery process and reduce throughput and packet transfer rates [83].

In a delay-tolerant network, an attacker lures packets by claiming to help them forward to their destinations. The adversary then selectively forwards packets to

cover up their blackhole or gray hole attacks by dropping packets [84].

- *Sybil Attacks:* For Sybil attacks, attackers send a message with multiple peer identities to control a part of UAV networks. Sybil attacks include three dimensions: communication, participation, and identification.

Communication dimension attacks establish connections between Sybil nodes and legitimate nodes in networks. Then, an attacker communicates with legitimate nodes through the connections to achieve interference effects on networks.

Participatory dimension attack means that attackers introduce Sybil nodes into the network in two ways, simultaneously or not simultaneously, so as to gradually control the network.

Identity dimension attacks refer to the identity spoofing of Sybil nodes. Attackers can assume identity through two alternative methods. One way is to steal the identity of a legitimate node, which is in a state where the network is disconnected or the battery is completely depleted. Another way is to create a new set of fake identities that do not exist in the network [85].

- *Hijacking Attacks:* UAVs rely on the video captured by a camera for target tracking and collision avoidance. The process is that flight controllers request the captured videos from the kernel of the computer operating system by issuing a system call [86]. When an aggressor knows the system parameters and can access the flight controller, the attacker could launch a hijacking attack on the system call, replace a real camera with a virtual camera, and set the UAV outside the original target, thus deliberately hijacking UAVs [33].

Information exchange between UAVs and GCS is easily hijacked, so UAVs could be accessed by unauthorized users [33]. The United States harvesters and predator UAVs, which are the most combative in Iraq, have been passively intercepted and attacked. Iraqi

militants intercept the captured real-time video source sent by UAVs to GCSs through SkyGrabber, which is an off-the-shelf cheap product [87].

SUMMARY OF CYBER DOMAIN ATTACKS

With the increase of UAVs, attackers are more willing to use these invisible attacks (such as cyber attacks) to disrupt civilian UAVs on missions or disrupt military UAVs on surveillance. As a result, UAVs are increasingly exposed to attacks in cyber domain.

Above all, in terms of traditional software threats, the threats to drones mainly come from keyboard Trojans, zero-day attacks, and a series of malicious software. Malware can break the confidentiality of UAVs or even destroy their usability, which may cause a serious failure for UAVs.

Second, as the rapid development of intelligent systems, intelligent systems related to UAVs have also been upgraded. Therefore, intelligent systems also face some security threats, such as model inversion attacks, model extraction attacks, poisoning attacks, and adversarial attacks.

Third, the network threats faced by UAVs are endless too, including injection attacks, fabrication attacks, DoS attacks, network eavesdropping attacks, man-in-the-middle attacks, etc.

In addition to this, UAVs could encounter other cyber attacks, such as stepping stone attacks that are launched against UAV networks. An attacker is interconnected through mobile hotspots and sends attack commands directly to all connected UAVs upon request. After UAVs accept the order from the attacker, the attacker begins to carry out targeted follow-up attacks on the UAVs [67].

UAVS SECURITY THREATS IN CYBER-PHYSICAL DOMAIN

A growing number of invisible threats have harmful effects on physical environment, such as the ability to disrupt cameras, turn off building lights, deflect cars off the road, or drop UAVs land in enemy hands. These kinds of invisible threats are cyber-physical attacks. They are implemented through the cyber space security holes to affect the physical environment [88]. In essence, cyber-physical attacks are a progressive alternative to physical attacks in crime, war, and terrorism. Cyber-physical attacks against UAVs mainly destroy the chips and sensors carried by UAVs to make UAVs vulnerable to cyber attacks such as spoofing attacks and jamming attacks. Such attacks are more appropriate for military UAVs, which carry out target tracking and reconnaissance [89].

- **GPS Spoofing:** In UAVs, GPSs are vital for UAVs. The high-precision position information provided by GPSs reduces or even eliminates the cumulative error of the inertial measurement units (IMUs).

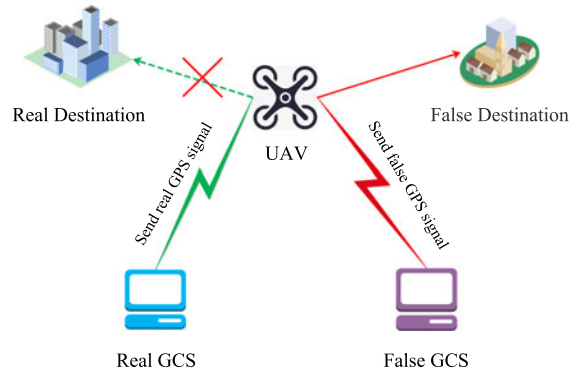


Figure 14.
GPS spoofing.

Under the combined operation of GPS and IMU, the navigation accuracy, reliability, and anti-interference ability of UAVs have been improved [4].

With the widespread application of GPS technology, security threats to GPS have gradually increased. As shown in Figure 14, GPS spoofing attacks falsify or replay GPS signals so that GPS receivers may calculate a wrong position and time information [68]. As a result, an attacker can conduct GPS spoofing attacks to UAVs such that the GCS cannot determine the exact location of UAVs.

In December 2011, the Iranian military captured a military UAV using GPS deception attacks. Iranian hackers used jamming attacks to inject interference signals between controllers and UAVs, causing operators to lose control of UAVs [32].

Some researchers conduct GPS spoofing experiments. He et al. [77] forged location to generate developed signals and use available radio peripherals to broadcast them. Once a UAV receives the wrong signal through the broadcast signal, it will get the wrong location and time information, causing all GPS function to fail. Finally, the attacker can control UAVs [90].

GPS spoofing attacks can not only use GPS signal generators to produce spoofed signals of the original products, but also realize GPS spoofing by embedding GPS malware into GPS receivers [33].

- **Optical Flow Spoofing:** To reduce the sensor drift effect when a UAV is not in use, the optical flow sensor of the UAV is equipped with GPS. A UAV can distinguish whether it is drifting by comparing the adjacent pixels of an image [91]. As UAVs are susceptible to sensor drift, attackers use this influence to launch spoofing attacks on UAVs.

Davidson et al. [92] conducted two spoof attack experiments, namely image spoof and laser spoof. In the image spoof experiment, they construct an image with a large number of corners and use a projector to shine the image on the ground. In the laser

spoof experiment, they point the laser to the ground to sense features. Comparing the results of these two experiments, they find that in an environment with fewer surface features, image deception for UAV is ineffective, while laser deception is effective. As a result, UAVs are more susceptible to laser spoofing and are more easily controlled in environments such as tiles, blankets, and concrete floors. The main reason why UAVs suffer from this attack is that the sensors that they carry cannot distinguish whether these surface features come from the enemy or the target ground.

There are also some sensors and technologies that can effectively suppress drift. For example, several types of DJI UAVs are equipped with binocular vision sensors with binocular vision ranging system technology, which can obtain exact location in space [91].

- *Other Sensor Attacks:* UAVs rely on gyroscopes to maintain balance. Son [93] introduced a method of attacking micro-electro-mechanical systems (MEMS) gyroscopes with ultrasound. Ultrasonic waves of certain frequencies could cause the gyroscope to resonate, thereby destroying the gyroscope.

UAVs use compasses for orientation, which are susceptible to electromagnetic interference. Once compasses are attacked, UAVs cannot tell the directions. Infrared sensors on UAVs are usually used to detect obstacles, but they are susceptible to light and blocks at close range. The sensor cannot work properly once affected by light and nearby barriers [91].

- *Summary of Cyber-physical Domain Attacks:* According to Table 3, there is a variety of attacks in the cyber-physical domain. This section introduces attacks in the field of cyber-physics, mainly attacking sensors of UAVs, such as GPS, infrared sensors, and optical flow sensors. Attacks on GPS include GPS jamming and GPS spoofing. However, many defense methods could be used against them, such as machine learning-based methods, cryptography-based methods, and spatial processing-based methods. It is worth noting that fewer defense methods focused on other sensor attacks, which may be a future research direction.

DEFENSE AGAINST VARIOUS ATTACKS OF UAVS

This section summarizes the corresponding defense methods according to UAV security threats in the section “UAVs Security Threats.” Since military UAVs are not available, most of the experiments and tests about defense methods are done with civilian UAVs, but these methods can be considered for use in military UAVs.

Table 3.

Summary of Cyber-Physical Domain Attacks	
Types	Description
GPS Spoofing	By forging or replaying the GPS signal, the GPS receiver calculates the wrong position and time information [66].
Optical flow spoofing	In environments with few surface features, drones are more susceptible to optical flow deception [89].
Other sensor attacks	Ultrasonic interference to the gyroscope, electromagnetic interference to the compass [89].

DEFENSE REGARDING PHYSICAL DOMAIN ATTACKS

The integrity and authenticity of hardware components of UAV systems should be guaranteed. Detection methods against malicious hardware attacks are as follows:

Conventional side-channel analysis, which includes timing and power analysis, can detect hardware Trojan activation. Since hardware Trojans usually change the parameter characteristics of the system, such as its expected performance and power consumption, this method can detect hardware Trojans [94].

Liu et al. [95] proposed statistical side-channel fingerprint identification methods to detect hardware Trojans. This method is also based on side-channel analysis, and its detection principle is to collect a signature of the parameters of the available chips and compare it with a trusted region in the multidimensional space, which needs to be established statistically. The final judgment result of this method is that the fingerprints of Trojan-free chips are expected to fall into the trusted region. In contrast, the fingerprints of Trojan-infected chips are expected to fall outside the region.

Thus, side-channel technology in the detection hardware Trojans can play an effective role. However, in most resource-constrained and technical situations, this approach is somewhat difficult to implement, because it depends on the details of the algorithm. With the improvement of the technology level of integrated circuits, the implementation difficulty of this method will be increased.

The defense methods against human factor attacks by UAVs are as follows:

Electronic antitheft locks can be used to reduce the probability of theft of UAVs. These antitheft locks can only be activated when they are actually located near GCSs or UAVs. At present, such electronic antitheft locks have been widely used in automobiles with smart

Table 4.

Summary of Traditional Software Threats Defense Methods		
Attacks	Proposed Solutions for Detection	Proposed Solutions for Defense
Malware	Malware analysis application [96]	Antivirus and firewall software
Keyloggers	Malware analysis application [96]	Use OS security level [95] Add sensitive information to whitelist [95]
Zero-day Attacks	Security Risk Framework [78] Use System Operator Agreement [77] Vulnerability identification method based on machine learning [97] Vulnerability detection method based on natural language processing [98] Use a static analysis tool to detect vulnerabilities [99] Use an enhanced greybox fuzzer to mine new vulnerabilities [100] Use a fuzzing framework to mine vulnerabilities [101]	N/A

keys [33], but electronic antitheft technology for UAVs is still scarce. Therefore, one of the future research directions is to strengthen UAV antitheft technology, such as adding electronic antitheft function.

The solution for UAV-friendly collisions is to reduce the possibility of UAV collisions by relying on Traffic Alert and Collision Avoidance System. Since the system can only be used by pilots, this method is mostly used for manned aircrafts, and it is also considered to be used in UAVs in the future [33].

DEFENSE REGARDING CYBER DOMAIN ATTACKS

Cyber domain attacks on UAVs are divided into traditional software threats, machine learning security threats, and

network threats. This section introduces the corresponding defense and detection methods against various cyber domain attacks, and then summarizes them in Tables 4–6.

DEFENSE AGAINST TRADITIONAL SOFTWARE THREATS

- *Malware Threats Defense and Detection Methods:* Malware is a typical software threat that can cause serious consequences if not detected immediately. Thus, the defense against malware is summarized as follows:

The simplest and earliest method is mainly to use antivirus and firewall software to protect software security, but the firewall software cannot prevent malicious knowledge, and cannot be put all threats such as Trojan.

Table 5.

Summary of Defense Methods for Security Threats of Machine Learning		
Attacks	Proposed Solutions for Detection	Proposed Solutions for Defense
Model inversion or extraction attacks	N/A	Differential Privacy Method [46], [47] Homomorphic encryption algorithm [54]
Poisoning attacks	Use filtering algorithm to detect [114]	Data source model based on improving data credibility [113] Use robust machine learning algorithms [88], [115]
Adversarial Attacks	N/A	Do a lot of confrontation training [69] Use more robust area-based classification [76] Use reverse cross-entropy defense method [156] Automate confrontation using tools such as neural networks [42], [43]

Table 6.

Summary of Network Threats Defense Method		
Attacks	Proposed Solutions for Detection	Proposed Solutions for Defense
Infection Attacks	Detection method based on machine learning [116] Message injection detection method [59] Detection method based on coding technology [60]	N/A
Fabrication Attacks	N/A	Defense method based on blockchain technology [118], [119], [121]
DoS Attacks	N/A	Defense method based on blockchain technology [121], [122], [123]
Jamming Attacks	IDS based on multichannel ratio (MCR) changes [34] IDS based on self-taught learning (STL) [128]	Methods based on statistical model [124], [125] Methods based on machine learning [126] UAV cybersecurity analysis simulation test platform [71] Methods based on Q-learning [127]
Network Eavesdropping Attacks	N/A	Use encryption technology [32] Use spread spectrum technology [129] Anti-eavesdropping device based on physical layer security [130] Privacy protection scheme [131]
Replay Attacks	Estimate the time and position using the shadows in the feed [77] match the control commands [77]	Define message verification parameters and define checksum by hash [132]
Man-in-the-middle Attacks	N/A	Hidden channel [133] Enhanced authentication protocol [134]
Wormhole Attacks and Blackhole Attacks	Adaptive IDS based on device specifications [135] Collaborative IDS based on machine learning [136] Centralized intrusion monitoring [66]	Enhanced AODV attack protocol [140]
Sybil Attacks	N/A	Symmetric key encryption [63] Consortium blockchain [141]
Hijacking Attacks	UAV honeypot [142], [143]	Network model based on encrypted channel [144]

Aviv et al. [96] suggested that all software, which can access sensitive information, should be checked before the software is put on the market to avoid the appearance of keyloggers. However, this method is not applicable to large-scale software.

Simon et al. [97] suggested that the OS security level can relieve all applications. For example, when an attacker wants to collect a password from a user, the OS can deny the request of the attacker to access sensitive data. If users still need to access

some sensitive information, the authors suggest to add the sensitive information into a whitelist.

Cai et al. [98] suggested that other malware analysis applications can be used to check malicious access to sensitive information.

- *Zero-Day Vulnerability Threats Defense and Detection Methods:* Zero-day vulnerabilities are unpredictable attacks, but some security frameworks could be used to reduce the risk of such an attack or design some detection systems to mitigate such attacks.

Loukas et al. [80] proposed a security risk framework to mitigate zero-day vulnerability attacks. Krishna et al. [79] stated that this attack can be mitigated by the use of redundant and well-organized system operator agreements to detect abnormal behavior.

Yamaguchi et al. [99] proposed an automatic identification method for code leak miss detection. This method is conducted using machine learning methods for analysis through an abstract syntax graph and a bag of words model. This method can identify the authenticity of missed inspections and find several zero-day vulnerabilities accurately.

Medeiros et al. [100] explored a new approach to detect vulnerabilities inspired by natural language processing (NLP). A sequence model is used to characterize vulnerabilities based on an annotated corpus. The model can be used as a static analysis tool to discover zero-day vulnerabilities in source code.

Cui et al. [101] proposed a static analysis tool to detect vulnerabilities based on graph comparison at the granularity of function. A weighted feature graph model is built to characterize function code syntactically and semantically in varying degrees of vulnerability sensitivity. This method is proved to be useful in detecting vulnerabilities.

Liang et al. [102] presented an enhanced gray-box fuzzer, which combines qualified seed generation, balanced seed selection, and hybrid seed mutation to explore deep paths as widely and as fast as possible. This method outperforms other commonly used fuzzers, and it is capable of triggering more crashes.

Yu et al. [103] constructed a fuzzing framework to mine vulnerabilities, which uses coverage-guided generative adversarial networks to generate test cases with high passing rates and code coverage.

To sum up, in order to prevent zero-day vulnerabilities, in addition to updating software patches in real time and repairing vulnerabilities, some algorithms should be combined to strengthen the construction of network intrusion prevention systems.

DEFENSE AGAINST MACHINE LEARNING SECURITY THREATS

- *Defense and Detection Methods for Model Inversion Attacks and Model Extraction Attacks:* The differential privacy method ensures that a model is trained with two different versions of a dataset. There can be no obvious statistical differences in the output of the model [104], [105]. Its purpose is to allow researchers to analyze a dataset without leaking information from a single sample [106]. The weakness of this method is the need to add a lot of randomization in the query results, which leads to a decline in the

availability of data. However, as an effective mathematical tool, differential privacy points out a development direction for privacy research.

A homomorphic encryption algorithm is a defense method based on the test stage [106]. This algorithm aims to allow the neural network to process the data without decrypting it, thereby protecting the confidentiality of a single input. However, its limitation is that the overhead is too large, and homomorphic encryption has additional constraints on the target deep learning model.

In [107], researchers improve a machine learning model to protect the privacy of training data and propose a method to provide a strong privacy guarantee for the training data. This method first learns a set of models trained with different sensitive datasets and uses them as teacher models. Then, the teacher model predicts the published unlabeled nonsensitive data and uses them to train student models. The advantage of this method is that it can ensure that the student model does not depend on any sensitive training dataset to resist the attacker from obtaining sensitive data. However, the disadvantages of this method are obvious. The early training process of this method is complicated and the time complexity is large.

- *Poisoning Attacks Defense and Detection Methods:* Since most poisoning attacks focus on attacking data and algorithms, the defense method against this attack starts with protecting data and algorithms.

Data protection refers to protecting the collected data from tampering, preventing injection, preventing data forgery, and detecting toxic data [108], [109]. Olufowobi et al. [110] proposed a data source model for IoT to improve the credibility and reliability of data. Baracaldo et al. [111] used source information as part of a filtering algorithm to detect poisoning attacks. This detection method could improve the detection rate.

Since learning algorithms are inherently uncertain, the attacker can poison these algorithms with this weakness. The most effective method against such attacks is to study robust machine learning algorithms. Candè et al. [112] investigated a robust machine learning algorithm. Goodfellow et al. [113] proposed a robust linear regression method that combines full low-rank matrix approximation with robust principal components to obtain a strong performance guarantee.

In summary, most defense mechanisms against poison attacks rely on the fact that poison samples are usually outside the expected input distribution. Therefore, poisoned samples can be treated as outliers, and existing algorithms can be used for data cleaning and robust learning to purify training samples.

- *Adversarial Attacks Defense and Detection Methods:* The defense methods against adversarial attacks mainly have two goals, namely preventing the generation of adversarial samples and detecting the adversarial samples.

At first, a large amount of confrontation training can be used to prevent some confrontation attacks. Huang et al. [114] proposed an earlier defense method, that is, by generating adversarial examples (AEs) as an intermediate step to learn a robust classifier with strong opponents. At the same time, they formulate a new search method for AEs.

Second, understanding the nature of the adversarial sample area and using a more robust area-based classification can resist adversarial attacks. Cao et al. [115] used an area-based classification instead of point-based classification to develop new DNNs. They predict labels by randomly selecting a few points from a hypercube centered on the test sample. Pang et al. [116] used a reverse cross-entropy defense method that maps standard samples to the low-dimensional manifold neighborhood of the final hidden layer space.

A number of studies use tools such as neural networks to counteract AEs automatically. Gu et al. [90] proposed a deep contractive network (DCN) with the contractive autoencoders and the denoising autoencoders, which can remove a lot of noise through additional noise corrosion and preprocessing adversarial noise. Akhtar et al. [117] reported a perturbation rectification network as the pre-input layer of the target model for combating the universal adversarial perturbations. The advantage of this method is that it can provide a defense to the deployed network without modifying the network and resist invisible hostile interference.

In short, the current defense technology against adversarial attacks mainly fall into three directions: changing the training process or input data, modifying the networks, and using additional networks. There are more and more studies in these three directions, and they are the hot research topics at the moment. Therefore, real-time and fast networks can be considered to improve the detection and defense about adversarial attacks.

DEFENSE AGAINST NETWORK THREATS

- *Infection Attacks Defense and Detection Methods:* Leonardi et al. [118] proposed a detection method based on neural networks. This method first distinguishes the types of transponders that use the carrier phase mode as a signal feature to classify the aircraft. Then, an algorithm is used to identify the received signal and determine whether the source of

the signal is legitimate. This method takes the carrier phase mode as the feature for detection. Its advantages are high accuracy, long action distance, and wide application. However, its disadvantages are limited by random errors and bias errors. In addition, due to the need to accept radio waves, it may be accessible to interference.

Habler et al. [61] proposed a message injection detection method that mainly uses a long-term, short-term memory codec algorithm to analyze a legitimate ADS-B message sequence to simulate a flight path. The aircraft can then autonomously evaluate the received ADS-B message based on these simulated routes and compare it with the previous legal approach to identify its deviation. This method is a typical anomaly detection for time series. However, since it is challenging to obtain abnormal data in the airspace at present, this method can be considered to be applied to actual scenes in the future.

Ramdhan et al. [62] proposed a detection method for detecting packet injection attacks. The principle of this method is as follows: when the receiving end decodes the correctly delivers data packet, the data packet is not subject to malicious injection attacks. Otherwise, it is subject to injection attacks. The optical coding operation can improve the security level of the transmission traffic. Therefore, it is recommended to use enough optical coding to encode the transmitted data packets in the free space optical communication link to improve the detection of injection attacks.

- *Fabrication Attacks Defense and Detection Methods:* UAVs should ensure the authenticity and integrity of the remote information. The following defense methods are mainly based on the blockchain technology:

Huh et al. [119] proposed an Internet of Things (IoT) application platform based on the blockchain technology to resist obstacles to fabrication attacks. This application platform uses the RSA public key cryptosystems, where the public key is stored in the blockchain, and the private key is stored on a single device. This platform is suitable for fabrication attacks suffered by the IoTs, which is a good research idea to be considered for application to UAVs.

Fan et al. [120] proposed a blockchain-based solution to solve the privacy problem in content-centric 5G mobile networks. This solution provides high-quality privacy protection through data encryption, access control, and key management to prevent message fabrication attacks. Although this scheme is not currently applied to UAV defense, future research can make efforts in this regard.

Ahmadi et al. [121] discussed a method of using blockchain technology to enhance network resilience. Meanwhile, they proposed a blockchain-based system to protect networks from fabrication attacks.

At present, in the field of UAVs, blockchain technology can assist traffic management, solve the problem of data storage, and solve the problem of trust crisis of Big Data in the UAV cloud system. However, there are few studies on using blockchain technology to solve the problem of UAV cybersecurity. Therefore, blockchain technology has a good research prospect in the field of UAV cybersecurity.

- *DoS Attacks Defense and Detection Methods:* When ensuring the availability of UAVs, it is mainly to detect or defend against DoS attacks. The detection methods for this attack are as follows:

Han et al. [122] proposed an intrusion detection system that distinguishes normal communication from communication generated by DoS attacks, thereby detecting whether a UAV is subjected to a DoS attack. Although the intrusion detection method has certain effectiveness, its most significant weakness lies in the detection speed. With the rapid growth of network data and the continuous development of network transmission tools, the processing speed of intrusion detection should be improved. Otherwise, it will lead to harmful situations such as packet loss and missing reports, which will further affect the accuracy and effectiveness of intrusion detection systems. Therefore, how to improve the detection speed of intrusion detection can be a central research direction in the future.

Dasu et al. [123] discussed the UAV air traffic control mode and proposed a distributed system based on regional chains. This system decentralizes UAVs and it can resist some DoS attacks without a central control.

In the same work, they also introduce two methods of preventing denial-of-airspace attacks. One is that the operators of UAVs can use cryptocurrencies on the blockchain to buy access to airspace, from municipalities to limit the overuse of other flights. The other would be for aviation regulators to reduce such attacks by using the transparency and immutability of the blockchain to limit space allocation for each registered UAV.

- *Jamming Attacks Defense and Detection Methods:* The defense methods of jamming attacks are as follows:

Some researchers [126], [127] proposed a method based on cognitive radio networks to reduce the impact on the system by improving the reliability of UAV networks. Experiments have proved the cognitive radio method proposed in this article to improve the reliability of wireless communication

of UAVs. However, this technology still faces many challenges from concept to application, especially many vital technologies that need to be breakthrough, making it become the research hotspot of UAV wireless communication in recent years.

Johansson et al. [128] proposed a machine learning method to help UAVs find a flight path on their own if they are disturbed and out of control. Its purpose is to ensure that UAVs can communicate with GCSs in the event of jamming attacks. The advantage of this method lies in its effectiveness, and it can automatically plan the optimal flight path. However, the experiment lacks actual data. Thus, it can consider applying in real scenarios in the future.

Javaid et al. [70] proposed a simulation test platform UAVSim, which allows users to specify different drone models and different attacks, then generate results graphically. They mainly launch jamming attacks on UAVs to collect relevant data and use these data to analyze the applicability of this simulation platform in detail. The final result shows that UAV-Sim can be used to evaluate attacks and mitigate various measures.

Min et al. [129] proposed a method of UAV defense based on the Q-learning. Although there are more and more defense methods based on reinforcement learning technology, the advantage of this defensive scheme is that the optimal defense performance of the defense system can be achieved without understanding the attack model. The defense system is based on the previous attack model and target.

Sliti et al. [36] proposed a jamming attack detection technology based on MCR changes. The advantage of this detection method is that it can recover the transmitted signal even in the presence of interference. Its basic principle is that when the received signal amplitude has obvious deviation, the receiver can detect interference transmission. When this deviation exceeds the threshold, a jamming attack will be detected. Then to eliminate the interference of the transmitted signal, the receiver records the received interference signal to calculate the interference MCR value. The calculated MCR value will be used to remove the interfering signal from the transmitter signal samples.

Hacking UAVs starts from two stages: interfering with the legal signal of UAVs, and sending a spoofing signal to a UAV to access its flight control device. So far, Arthur [130] proposed an adaptive IDS based on deep learning algorithms to identify intruders of drones and ensure UAVs safe return-to-home (RTH). The deep learning algorithm improves the accuracy of IDS in detecting unknown attacks from hostile theaters, facilitates learning powerful functions from unmarked data of unknown attacks, and provides

access to continuous and huge data sets. They add a self-taught learning (STL) algorithm, which performs best on unlabeled data.

- *Network Eavesdropping Defense and Detection Methods:* Network eavesdropping is the act of listening transmissions, which is the most direct security threat in any network. There are several ways to mitigate the effects of eavesdropping:

The use of encryption techniques can protect the confidentiality of the information and the transmission source authentication effectively. The encryption method is divided into two schemes: symmetric and asymmetric. In the former, encryption and decryption are performed by their unique keys. The latter, each user has its pair of public-private keys. Public key encryption and private key decryption can protect data integrity and confidentiality [38]. Encryption technology is an effective means of protecting against network monitoring attacks, but it is also applied to electronic commerce VPN and IC cards. Therefore, future research can consider using this technology in UAV network security, such as the identity verification of UAVs.

Spread spectrum technology can be used to reduce the effects of listening. It is difficult for an attacker to monitor a situation regardless of whether he/she knows the spreading code or frequency hopping pattern or not, because the transmitter and receiver both need to use a known spreading code or frequency hopping pattern. But these two methods do not apply to broadcast characteristics having GPS and ADS-B [131].

Shang et al. [132] described a technique based on physical layer security antieavesdropping device. The device is mainly directed to the beam and transmits bug artificial noise signal interference active eavesdropper, and it is mainly suitable for vehicle-to-everything(V2X) communications, and UAVs are also included in V2X. Although this device acts as a UAV, after the deep research, this device can be used in UAV systems.

Zhang et al. [133] proposed a privacy protection scheme without the pairing of vehicle networks and 5G smart grid slices. This method involves a Paillier encryption system and Hash-then-Homomorphic technology. It can ensure the security and privacy of V2V communication, and effectively prevent network eavesdropping attacks, and can be applied to UAV networks.

In a word, the defense methods against network eavesdropping mostly use traditional methods to resist. Therefore, future research can consider adopting new intelligent methods for defense or detection.

- *Replay Attacks Defense and Detection Methods:* A detection method is to estimate the time and position on the operator side using the shadows in the feed and compare them to the expected data. Another method matches the optical flow of a UAV fed from the camera with the control command from the operation control unit [79].

Haddadpajouh et al. [134] proposed a defensive method, which defines the time stamp and identity verification parameters to effectively defend against replay attacks.

- *Man-in-the-Middle Attacks Defense and Detection Methods:* Man-in-the-middle attacks are common means of information domain attacks. The defense against these attacks are as follows:

The DoD defines a covert channel as “any communication channel that a process can use to transmit information in a manner that violates the system’s security policies.” Taylor et al. [135] proposed that hidden timing channels and storage channels can be used to provide some level of identity verification. Messages sent from one UAV to another on the network are legal and can be executed. Modbus Covert Channel Integrity Check can pass data that can verify the integrity of the message, and attackers cannot be aware of the transmission. Therefore, this defense method can be considered to be applied to UAVs.

Azad et al. [136] developed a new type of self-enhanced authentication protocol for next-generation networks based on the session initialization protocol. The protocol is based on low-entropy shared passwords without relying on public key infrastructure (PKI) or trusted third-party systems. This method enables proxy servers and session initiation protocol clients to exchange their authentication information over open and insecure networks based on passwords, without any PKI. Thus, it can be considered to be applied to the defense of UAVs.

- *Wormhole Attacks and Blackhole Attacks Defense and Detection Methods:* For this type of security threats, most researchers use a variety of IDSs to detect vulnerabilities, and also protect the UAV systems using some of the security frameworks.

Mitchell et al. [137] proposed an adaptive IDS based on device specifications. This system can detect suspicious UAVs in cooperative operations under the condition of strong cooperation continuity, and can evaluate UAVs under malicious attacks quickly.

Abusitta et al. [138] proposed a collaborative IDS based on machine learning, which can use the data provided by historical records to make active decisions effectively on any intrusive behavior. This

system uses a noise reduction autoencoder as the construction fast to build a deep neural network.

After a black hole or gray hole attack occurs, all or parts of the received data packets are discarded. Therefore, IDSs for this attack are to start centralized intrusion monitoring.

IDS is a kind of network security equipment, which can monitor network transmission in time. It will alarm or take active response measures when it finds that the network transmission can carry out. Also, IDS is different from other network security devices in that IDS is a proactive security protection technology. However, its disadvantage is that the false positive rate is high. Therefore, how to reduce the false positive rate of IDS can be regarded as a direction of future research.

Sedjelmaci et al. [68] introduced a detection strategy that all UAVs near GCSs send an adjacent data packet to GCSs. This data packet includes the node type, the next-hop node, and the previous-hop node. First, the node type refers to whether the node is the source node or the relay node. Second, the next-hop node refers to UAVs forwarding the data packet to the adjacent Node. Finally, the last hop node refers to the adjacent node, where a UAV receives the data packet. To reduce false positives, GCSs collect adjacent data packets from normal and suspicious UAVs, and ignore data packets received from abnormal and malicious UAVs. Then, GCSs check whether the relay node forwards the data packet and calculates the number of dropped data packets. Finally, it decides whether it is attacked by a black-hole or a gray hole.

Mobile AdHoc network (MANET) is the network structure where the current UAV is located, but it is vulnerable to attacks. The temporary on-demand distance vector routing (AODV) protocol is one of the targets of blackhole attacks. Therefore, in [139], [140], [141], many experts focus on the simulation experiments of blackhole attacks against AODV protocol, so as to analyze the influence of these attacks on AODV. However, since most existing work requires higher computing resources and the MANET is very limited in some cases, lightweight security enhancements to the AODV protocol are required. Furthermore, Fu et al. [142] proposed an AODV attack protocol to reduce the possibility of blackhole attacks.

- *Sybil Attacks Defense and Detection Methods:* Vasudeva et al. [85] proposed a solution based on a symmetric key cryptography method that resembles the Needham-Schroeder protocol. The protocol uses a trusted third party or base station, and then generates and distributes unique keys before all nodes are deployed to the network. The base station retains the data of all the nodes, and then uses this data to authenticate the nodes, and generates a secret session key for

any pair of nodes to establish a secure connection. This method has a certain degree of confidentiality, but it needs to know the encryption protocol in advance, so the universality is weak.

Mehta et al. [143] recommended using the consortium blockchain in the network because it is more secure than the public blockchain and can resist Sybil attacks. This work also provides other methods to prevent Sybil attacks, such as assigning different permissions to each participating member in the same network.

- *Hijacking Attacks Defense and Detection Methods:* Nawrocki et al. [144] launched the concept of honeypots. The core concept is that honeypots are deceptive computer resources, which can be used to probe, attack, or compromise. As honeypots are very fragile, they are often vulnerable to attacks, which is an effective way to sense threats. Based on this feature, Vasilomanolakis et al. [145] applied honeypots to UAV defense. When the UAV honeypot signals are stronger than the actual UAV, and are placed in a strategic position, they can be used to detect and mitigate UAV attacks.

In order to solve the hijacking attacks on UAVs, Yoon et al. [146] proposed a high-level idea of encrypted channels and design a network model based on this idea. This model includes GCSs, middleware, and UAVs. It can deal with network security issues, such as DoS attacks. The defense process is to manage two communication channels through middleware and detect the intrusion of anonymous attackers on the main channels. If the middleware detects an intrusion, it will conduct a DoS attack on the main channel, causing the attacker to lose control of the main channel, thereby preventing data from leaking through the hijacking channel. The specific detection method of this model is to handle the authentication by using encrypted index values in the communication channel, and prevent attackers from invading.

DEFENSE REGARDING CYBER-PHYSICAL DOMAIN ATTACKS

Cyber-physical domain attacks are mainly aimed at UAV sensors; this section reviews GPS spoofing and MEMS gyroscope defense and detection methods and summarizes them in Table 7.

- *GPS Spoofing Defense and Detection Methods:* Although cryptography is difficult to defend against GPS spoofing, there is still a breakthrough in the detection of this attack. Wesson et al. [147] described a technique based on probabilistic models and statistical hypothesis testing. This technique is mainly to

Table 7.

Summary of Cyber-Physical Defense Method		
Attacks	Proposed Solutions for Detection	Proposed Solutions for Defense
GPS Spoofing	Probability model and statistical hypothesis testing [145] Asymmetric encryption method [146] Machine learning methods [148], [149] Behavior analysis and monitoring system [27] [152]	Phase delay measurement and spatial processing [147] Crowd-GPS-Sec method [150] Antenna array method [151] Vision and inertial navigation system [153]
Gyro Interference	N/A	Two physical isolation methods are foam [154] and nickel fiber protective cover [155]

verify the identity of GPS signals. Kerns et al. [148] reported a method based on asymmetric encryption, which mainly uses reserved and available bits in civilian navigation messages for periodic signatures. When the sender uses this periodic signature, the receiver verifies the signature and the GPS signal source to detect GPS spoofing attacks. Although the protection of GPS signals through encryption has played a specific role, the existing transmission protocol inevitably needs to be changed, so the feasibility is somewhat tricky.

Konovaltsev et al. [149] proposed a defense method based on phase delay measurement and spatial processing to protect the GPS signal receiver from spoofing attacks. Its workflow is to first perform phase delay estimation to detect the attack, and then use the antenna array to calculate the direction of arrival to determine the direction of GPS signal arrival and determine the spoofing signals in other directions. However, this method has a high demand on the antenna, and the detection effect is not stable.

Machine learning algorithms can be used to determine hidden patterns in a set of received GPS signal data and find their anomalies. Panice et al. [150] proposed a method based on state estimation analysis to detect GPS spoofing attacks on UAV systems by using support vector machines (SVM). Manesh et al. [151] proposed a machine learning method based on a supervised artificial neural network, which detects GPS attacks launched by recording and playing back GPS signals.

Combining several defense methods can improve the performance of the system against attacks. Jansen et al. [152] proposed the Crowd-GPS-Sec method that mainly uses wireless air traffic control systems such as ADS-B and traditional positioning methods such as TDoA, as well as position differences reported by

multiple aircraft to detect spoofing attacks at remote locations.

Ni et al. [153] proposed a defense method against GPS jamming attacks, through array antennas specially designed for aircraft. Once the direction of the interfering signal is found using this method, the array beam pointing at the source can be zeroed by orthogonal vector weighting to suppress the interfering signal.

Birnbaum et al. [32] elaborated upon a surveillance system using behavior analysis, which can be used to capture flight data and estimate the current behavior state, and continuously compare the estimated behavior with the initial behavior model. If a significant change in behavior is observed, the system will alert the ground control station and can perform predefined actions. The advantage of this system is that it does not rely on GPS to detect the deviation of the flight path, but uses a safer flight data source dedicated to UAV to detect the deviation of the same flight path. Finally, the experiment results show that this monitoring system can automatically determine the flight behavior of the UAV and detect deviations, and the behavior detector can successfully identify anomalies under various weather conditions without false alarms.

Birnbaum et al. [154] defined a novel UAV flight behavior method, and base on this method, a UAV surveillance system is studied. The system can capture flight data and estimate fuselage and controller parameters, and continuously compare the estimated parameters with previously known parameters. If a significant change in parameters is observed, the system may alert GCSs and can perform predefined operations. Finally, the experiment results show that this monitoring system can automatically establish the UAV flight parameters and detect obvious deviations.

The solution for GPS signal jamming is to use an

alternative method of navigation. Wu et al. [155] proposed a visual and inertial navigation system that enables UAVs to autonomously navigate when it cannot receive GPS signals.

- *Gyro Jamming Defense and Detection Methods*: Various types of sensors in UAVs, once attacked, may output erroneous data to UAVs, and even destroy the sensors. The biggest threat to the MEMS gyroscope is the resonance problem. Therefore, in order to solve this problem, researchers have successively proposed some physical isolation methods to reduce the noise interference received by the gyroscope. Roth [156] excavated a scheme for physically isolating MEMS sensors using foam, and experiments have proved that this method can greatly reduce the interference of ultrasonic waves on MEMS sensors. Soobramaney et al. [157] detailed a physical isolation method, which uses a protective cover made of nickel fiber to reduce the interference of ultrasonic noise on the MEMS gyroscope. Although these physical isolation methods can alleviate the interference of ultrasonic waves on the gyroscope, they also bring new problems, such as foam will cause poor heat dissipation.

FUTURE RESEARCH CHALLENGES

UAVs are widely applied in various fields, such as reconnaissance, strike, border patrol, precision irrigation, and express delivery, so they face many security threats. Therefore, the security of UAVs is extremely significant. Future research on UAV security can be carried out from the following aspects.

THE PROSPECT OF UAVS PHYSICAL DOMAIN RESEARCH

The physical security of large UAVs is mainly to strengthen the ability to avoid obstacles, so as to prevent the collision. Therefore, this article thinks that we can develop stronger materials and make the body of a UAV, to reduce the power of destructive attack, so as to play a certain protective role.

- *Improving the Physical Reliability of UAVs*: A UAV flight base station may encounter physical reliability issues independent of the existing ground base stations. For example, a UAV might accidentally run out of battery, lose control due to a sudden gust of wind, or even collide with another airborne object. The physical reliability issues ultimately affect the quality of the wireless service and the overall quality of the user experience. Therefore, UAV physical reliability must be fully solved, which can be

positioned as a future research direction. It may be a long time before we can expect to get highly reliable UAV systems at the lowest cost.

- *Establishing UAVs Early Warning Mechanism*: Small UAVs are easy targets at low altitudes. Therefore, it is necessary to develop a more robust body to withstand some destructive attacks, and to add some warning modes, so as to be able to sense the attack threat around them.
- *Enhancing UAV Perception and Collision Avoidance Technology*: In the face of the increasing number of UAVs, perception and collision avoidance technology is still one of the research priorities. Only with the continuous maturity of this technology can it promote the expansion of UAVs application field and ensure the flight safety of UAV or manned aircraft in airspace. Therefore, the functions in UAVs perception and collision avoidance technology can be enhanced and optimized, so as to improve its obstacle avoidance ability.
- *Improving Obstacle Avoidance Route Planning of UAVs*: Obstacle avoidance path planning remains a hot research topic in the field of UAVs, and it plays an important role in ensuring the flight safety of UAVs and improving the efficiency of their missions. In this regard, the following directions for future development are proposed:

First of all, the research on the 3D path planning in complex environment can be strengthened. Second, traditional planning methods should be combined with the new generation of artificial intelligence technology to complement each other's advantages. In this way, we can further solve the existing problems in the traditional planning methods and actively overcome the problems in the application of artificial intelligence methods in route planning. Finally, the actual performance and operational characteristics of onboard sensors should be fully considered in the future research of path planning methods.

THE PROSPECT OF UAVS CYBER DOMAIN RESEARCH

UAV security in the cyber domain is a need for all types of UAVs, because cyber attack is the most common security threat to UAVs. Nowadays, with the rise of technology of Unmanned Aircraft Systems Swarm (UAS Swarm), its biggest threat is network attacks. Therefore, how to design a new security and efficient routing protocol can be one of the research directions of UAV network security. Other research directions related to the security in the cyber domain are as follows.

- *Strengthen the Encryption of UAVs Communication Links:* In a UAV system, there are multiple communication links connected to UAV, including UAV to ground station, UAV to air traffic control (ATC), UAV to UAV, and UAV to external computing platform (for example, cloud computing and edge computing). Once these communication links are attacked, a UAV operator may lose control of UAVs or lose important data. At present, most studies on communication security focus on strengthening the communication protocol or encrypting communication links against an attack.
- *Establishing UAVs Communication Vulnerabilities Threat Identification Model:* Currently, vulnerabilities related to UAV communication are emerging in an endless stream. If a threat model for identifying various types of communication vulnerabilities in drones can be provided, it will greatly promote the research of drone communication security. This model can help users quickly understand where to check and how to find vulnerabilities in drone communications, so that they can better understand how to protect communications.
- *Design of a New Efficient and Secure Routing Protocol for UAVs:* Compared with traditional computer networks and wireless networks, mobility and mechanical degrees of freedom and their operating conditions make drone networks a separate category. This kind of network is the ad hoc network, which is characterized by its dynamic topology and no central base station constraints, and its routing is vulnerable to attacks. Therefore, designing a new safe and efficient routing protocol is the current research direction of UAV network security.
- *Building a UAV Detection and Defense Integrated Multilayer Security Framework:* Under weak network security protection, UAV networks may be subject to security threats such as unauthorized connection, illegal access, malicious control and attacks. Therefore, we can explore a methodology that provides unified protection for such networks, which is a multilayer security framework that includes detection and defense methods.
- *Performing Unencrypted Device Identification:* Nowadays, a typical cybersecurity threat is an identity spoofing attack, in which an attacker eavesdrops on existing wireless communications, and then impersonates the identity of a legitimate drone to send false messages or engage in other malicious activities. The existing solutions are based on cryptography, but this approach depends on the confidentiality of the encryption key, which faces many threats and challenges. Therefore, an unencrypted solution is required to

verify the identity of UAVs, thus ensuring the reliable and legitimate UAVs. It can be seen that recognition based on nonencrypted devices can also be a major research direction in the future.

- *Encryption for UAVs Privacy Data:* UAVs are used to perform data collection tasks in various situations, such as remote sensing, surveillance, and infrastructure inspection. These tasks are usually completed by different UAV sensors. The data collected may contain various sensitive information. For example, the surveillance video data recorded by the high-resolution camera carried by the UAV may be confidential. If these data are leaked, it may cause unpredictable consequences, and the main vulnerability that leads to the leakage of UAV data privacy is lack of identity verification or insufficient identity verification. Therefore, encryption of privacy data for UAVs is the focus of future research on data security of UAVs.
- *Developing Cooperative Countermeasures for Multiple UAVs:* Most existing countermeasure algorithms are limited to a single UAV system. While most of studies discuss countermeasure algorithms for multiple UAVs scenarios, they cannot provide any specific solutions. Therefore, the future research direction can be the development of cooperative countermeasure algorithms for multiple UAVs. For example, we can modify and adopt an existing distributed security framework, such as Kerberos, for multiple UAV systems.
- *Design of Light-Weight Artificial Intelligence Algorithms for UAVs:* Under the tendency of global development of IoTs, light-weight artificial intelligence algorithms are needed to respond quickly and accurately for devices with strict computing constraints, such as UAVs. Most studies assume that AI algorithms are executed on cloud servers, which have rich computing resources. However, resource constraints are often ignored in these studies. UAVs usually perform low-complexity algorithms, such as navigation and pathfinding, but they have no way to execute or train data on superior computing servers. Therefore, future research can consider designing light-weight AI algorithms for use in UAVs.

THE PROSPECT OF UAVS CYBER-PHYSICAL RESEARCH

Security issues in this domain are mainly focused on the sensors of UAVs. Sensors are an important source of the UAVs' ability to capture data from the outside world. Once the sensors are interfered or tricked, UAVs cannot acquire their current real status information, thus making a wrong response and rendering the UAV unable to carry out its normal

mission. However, the ability to detect security threats in UAV sensors is still weak. In future research, we should continue to uncover the security threats of sensors, so as to design corresponding algorithms to avoid the adverse effects brought by these security threats.

CONCLUSION

Due to the autonomous, flexible, easy to use, low cost, and low energy consumption nature of UAVs, they bring many economic, commercial, and industrial benefits to civilian and military fields. However, their extensive usages also bring many security problems. From the perspective of UAV cybersecurity, this article comprehensively summarizes the attack of UAVs and its corresponding defense means from the physical domain, cyber domain, and cyber-physical domain, respectively. Nowadays, most of UAV technologies combine with intelligent algorithms, so attacks against these algorithms can result in a breakdown of the intelligent systems of UAVs and make UAVs unable to function properly. Therefore, this article adds the classification of security threat of UAV intelligent system to summarize and analyze several types of attacks suffered by UAV intelligent systems and corresponding defense methods.

By comparing the existing UAV related surveys, we find that most reviews analyze from a single perspective and few reviews analyze and summarize UAV cybersecurity. Therefore, this article provides a comprehensive survey on the topic of UAV cybersecurity. First, we generalize several typical physical threats on UAVs, such as physical hard-kill and physical soft-kill attacks, hardware Trojan attacks, and human attacks. Then, we touch upon the defense methods of these attacks and present the corresponding future direction. Second, this article reviews all kinds of attacks on UAVs and corresponding defense methods, such as injection attacks, jamming attacks, DoS attacks, man-in-the-middle attacks, network eavesdropping attacks, etc. In particular, a highlight of this article is the addition of the section on security threats of intelligent systems, which includes model inversion attacks, model extraction attacks, poison attacks, and adversarial attacks. Finally, this tutorial surveys the attack and defense methods of various sensors of UAVs, such as GPS spoofing attacks. In short, we make a comprehensive review of the security threats and their opportunities and challenges for UAVs in the future.

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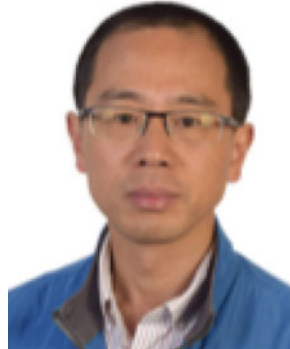
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Zhenhua Yu (Member, IEEE) is currently a Professor with the Institute of System Security and Control, College of Computer Science and Technology, Xi'an University of Science and Technology, Xi'an, China. His research mainly focuses on cyber-physical systems, unmanned aerial vehicles, multi-agent systems, artificial intelligence

system security, and nonlinear dynamical systems. He received the Ph.D. degree from Xi'an Jiaotong University, Xi'an, China, in 2006.



Zhuolin Wang is currently working toward the master's degree in computer technology from Xi'an University of Science and Technology, Xi'an, China. Her research interests include unmanned aerial vehicles, deep learning, and anomaly detection.



Jiahao Yu is currently a Mission Advisor at the Universal Weather and Aviation, Inc. He has experience in Part 91 Private and Part 135 Charter flight operations and has conducted research related to airport operations simulation, airport/aircraft evacuation simulation, data analysis, cybersecurity workforce training, simulation fidelity, etc.



Dahai Liu (Senior Member, IEEE) is currently serving on the editorial board for the *International Journal of Industrial Ergonomics*, *Human Factors and Ergonomics in Manufacturing and Service Industries*, and he has been an editorial member for the *International Journal of Aviation Psychology* from 2009 to 2019. His current research

interests include system engineering design and optimization, systems modeling and simulation, human factors engineering and artificial intelligence. He received the Ph.D. degree in industrial engineering from the University of Nebraska-Lincoln in August 2022. He is a senior member of IIE.



Houbing Herbert Song (Fellow, IEEE) is currently a Professor at the Department of Information Systems, University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA, and the Director of the Security and Optimization for Networked Globe Laboratory (SONG Lab). His research

interests include cyber-physical systems, cybersecurity and privacy, Internet of Things, edge computing, AI/machine learning, Big Data analytics, unmanned aircraft systems, connected vehicle, smart and connected health, and wireless communications and networking. He received the Ph.D. degree in electrical engineering from the University of Virginia, Charlottesville, VA, USA, in 2012.



Zhiwu Li (Fellow, IEEE) is currently with the Institute of Systems Engineering, Macau University of Science and Technology, Macau, China. His current research interests include the Petri net theory and application, supervisory control of discrete event systems, workflow modeling and analysis, system reconfiguration, game theory,

and data and process mining. He received the Ph.D. degree in manufacturing engineering from Xidian University, Xi'an, China, in 1995. He is a member of the Discrete Event Systems Technical Committee of the IEEE Systems, Man, and Cybernetics Society and a member of the IFAC Technical Committee on Discrete Event and Hybrid Systems from 2011 to 2014.